

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (EAT / UTC+3) · Remote

PROFESSIONAL SUMMARY

Senior full-stack engineer — **React** and **Node.js** — with 6+ years building type-safe services and production front ends for US, Singapore and East African teams, fully remote. Led the end-to-end migration of **20+ production applications** to Firebase/GCP with **zero downtime**, running a real-time two-way MongoDB↔Firestore sync over Node.js and Google Pub/Sub so both stacks served live traffic during cutover, and took **\$5,000/month** off the infrastructure bill. Architected containerized microservices with Apache Kafka, **Docker** and Kubernetes, with typed cross-service contracts over gRPC. Comfortable across the whole stack: React UIs from Figma under agency deadlines, Express/Cloud Functions APIs, and **cloud deployment workflows** on GCP and AWS. GitHub: github.com/zac-09

CORE COMPETENCIES

React

Node.js

TypeScript

Docker & Kubernetes

API Design & Typed Service Contracts

CI/CD & Zero-Downtime Deployment

Cloud Deployment (GCP, AWS)

Performance Optimization

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning **cloud deployment workflows** and release sequencing end to end, reporting directly to the CTO.
- Implemented real-time two-way sync between MongoDB and Firestore using **Node.js** and Google Pub/Sub, so both stacks served live production traffic in parallel throughout cutover.
- Saved the company **\$5,000/month** by migrating all services off AWS to GCP.
- Implemented Express Shipping end to end — backend, pricing and checkout surface — generating an additional **\$40 revenue per order**.
- Built a 3D visualisation feature using video overlay and the ffmpeg library, increasing buyer conversion.
- Migrated file storage from Amazon S3 to GCS via a Python script; migrated a WebFlow site from a Parse backend to Firebase.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore and GCP; **converted to a full-time role** on delivery.
- Built a migration pipeline reading from MongoDB and writing to Firestore after complex processing logic.
- Wrote engineering **documentation** for the new Firebase SDKs (Node.js and Python) and trained the Ops team on the Firebase Dashboard and backup procedures.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected a **containerized microservices** backend using Apache Kafka, **Docker** and Kubernetes; built typed cross-service contracts over **gRPC**.
- Led a team of 4 developers building legal tech systems for NGOs in Uganda.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.
- Developed a cross-platform mobile app for LASPNET using **React** Native and Expo (iOS & Android) with Firebase push notifications.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led **performance optimisation** of a PWA delivering real-time stock market prices from a Firebase backend.
- Built ad-hoc jobs querying over **2 million Firestore records**, keeping customer prices current from Morningstar APIs.
- Added responsive UI pages to the PWA using **React**, HTML and Tailwind; extended backend APIs using Firebase Cloud Functions and Express.js.

Software Engineer · US Remote

- Full-stack development for a US-based agency with demanding clients, reporting directly to the CTO.
- Built backends for DeFi applications using Web3 and **Node.js**.
- Delivered **React** UIs from Figma designs under tight timelines, collaborating directly with designers.

PROJECTS

Parse → Firebase Migration Zero downtime

Migrated 20+ production applications off Parse/MongoDB to Firebase/GCP without taking the business offline, sequencing app-by-app cutover with rollback at every step.

Node.js · Firebase · GCP · MongoDB

Real-time Pub/Sub Sync Pipeline Live traffic

Two-way MongoDB ↔ Firestore sync pipeline that let both stacks serve parallel production traffic during migration.

Node.js · Google Pub/Sub · Firestore

Healthcare WhatsApp Chatbot AI / OCR

Integrated OpenAI and Google Vision for prescription OCR delivered over the WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine Matching

Collaborative and content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, React, TypeScript, JavaScript, Firebase, GCP, Docker, Kubernetes, CI/CD, Python, MongoDB, SQL, HTML, CSS, Git, Linux

Intermediate: Go, AWS, Rust, React Native

Architecture: REST & typed RPC APIs (gRPC/protobuf), event-driven microservices (Apache Kafka, NATS Streaming, Pub/Sub), zero-downtime migration

Languages: English, Kiswahili